

Meeting Log — Smart Scientific Calculator

Instructions: Fill in the attendees, discussion points, and action items for each meeting. Add new entries as meetings occur.

Meeting 1

Field	Detail
Date	[Fill in: e.g. 12 Aug 2026]
Time	[Fill in: e.g. 2:00 PM – 3:00 PM]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Initial brainstorming and project-topic selection.
2. Discussion of feasible technology stacks.
3. Assignment of preliminary research tasks.

Discussion Summary

- The team decided to build a **Smart Scientific Calculator** as the project topic.
- Agreed on **HTML/CSS/JS** for the frontend and **Node.js + Express + MongoDB** for the backend.
- Discussed the incremental development model as the process framework.

Decisions Made

- Project topic finalised: Smart Scientific Calculator.
- Development model: Incremental.
- Technology stack: HTML, CSS, JS, Node.js, Express, MongoDB.

Action Items

Action	Owner	Deadline
Draft initial problem statement	[Member 1]	[Date]
Research similar calculator applications	[Member 2]	[Date]
Set up project repository	[Member 3]	[Date]

Action	Owner	Deadline
Create initial UI wireframe sketch	[Member 4]	[Date]

Meeting 2

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Review problem statement and objectives.
2. Define functional and non-functional requirements.
3. Plan Iteration 1 scope and task distribution.

Discussion Summary

- Reviewed and refined the problem statement.
- Identified six user needs (UN-01 to UN-06) to be addressed across six iterations.
- Finalised functional requirements: basic arithmetic, scientific functions, error handling, responsive UI, and database history.
- Finalised non-functional requirements: performance (< 1s response), usability, browser compatibility.

Decisions Made

- SRS document structure agreed upon.
- Iteration 1 scope: basic arithmetic operations (+, -, *, /, %, AC, DEL).

Action Items

Action	Owner	Deadline
Write SRS document	[Member 1], [Member 2]	[Date]
Design use-case diagram	[Member 2]	[Date]
Implement Iteration 1 (basic arithmetic)	[Member 1], [Member 3]	[Date]
Design CSS layout for calculator grid	[Member 4]	[Date]

Meeting 3

Field	Detail
Date	[Fill in]
Time	[Fill in]

Field	Detail
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iteration 1 (basic arithmetic).
2. Plan Iterations 2 & 3 (scientific functions).
3. Review architecture diagram.

Discussion Summary

- Iteration 1 was demonstrated successfully — all basic operations work.
- Discussed scientific function integration: `sqrt`, `sin`, `cos`, `x^y`.
- Reviewed the proposed 3-tier architecture (Frontend → Express Server → MongoDB).

Decisions Made

- Iteration 1 accepted.
- Scientific functions will use degree-based input (not radians).
- Architecture diagram approved.

Action Items

Action	Owner	Deadline
Implement <code>sqrt</code> , <code>sin</code> , <code>cos</code>	[Member 2]	[Date]
Implement <code>x^y</code> (power function)	[Member 4]	[Date]
Finalise architecture diagram for report	[Member 3]	[Date]

Meeting 4

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iterations 2 & 3 (scientific functions).
2. Plan Iteration 4 (error handling).
3. Discuss UI improvements for Iteration 5.

Discussion Summary

- Scientific functions demonstrated and working.
- Identified edge cases: division by zero, negative square root, malformed expressions.
- Discussed responsive grid layout and button feedback for better UX.

Decisions Made

- Iteration 4 to use try-catch blocks with user-friendly "Error" messages.
- Iteration 5 to use CSS Grid for the keypad and real-time display updates.

Action Items

Action	Owner	Deadline
Implement error handling (Iteration 4)	[Member 1]	[Date]
Redesign UI with CSS Grid (Iteration 5)	[Member 4]	[Date]
Begin database schema design	[Member 3]	[Date]

Meeting 5

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iterations 4 & 5.
2. Plan Iteration 6 (history & database).
3. Plan test-case design.

Discussion Summary

- Error handling and UI improvements demonstrated successfully.
- Discussed MongoDB schema: `{ expression, result, createdAt }`.
- Planned REST API endpoints: `POST /api/history`, `GET /api/history`.

Decisions Made

- Use Mongoose for schema validation and MongoDB interaction.
- History retrieval limited to 10 most recent records, sorted by `createdAt` descending.

Action Items

Action	Owner	Deadline
Implement Express server + MongoDB (Iter. 6)	[Member 3]	[Date]
Connect frontend to API	[Member 1]	[Date]
Design test cases for all iterations	[Member 2], [Member 4]	[Date]

Meeting 6

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Full integration demo (all 6 iterations).
2. Review test results.
3. Plan mid-semester submission packaging.

Discussion Summary

- Full end-to-end demo completed: basic arithmetic → scientific → error handling → UI → database history.
- All 19 test cases passed.
- Discussed mid-semester submission structure and report writing assignments.

Decisions Made

- Each member assigned specific report sections.
- Demonstration video to be recorded before submission deadline.

Action Items

Action	Owner	Deadline
Write report sections (assigned)	All members	1 Oct 2026
Record demonstration video	[Member 1], [Member 3]	2 Oct 2026
Package ZIP for submission	[Member 2]	3 Oct 2026
Final review and submission	All members	3 Oct 2026

Reminder: Replace all [Fill in] and [Member N] placeholders with actual dates and names before submission.

